An improved genetic algorithm for 2D triangular protein structure prediction

Shih-Chieh Su and Jyh-Jian Tsay
Department of Computer Science and Information Engineering
National Chung Cheng University, Taiwan
{ssc95p, tsay}@cs.ccu.edu.tw

Abstract

A great challenge in computational molecular biology is to predicate the protein structure from its primary amino acid sequence. Since the HP lattice model [was] developed by Lau and Dill [2] and used by others to derive similar approximation and heuristic search algorithms, a variety of lattice models have been proven useful to explore the relationship between the primary amino acid sequence and its native folding structure, particularly in protein folding problems (PFP) and protein structure prediction (PSP). Many researchers have favored and focused research on the 2D square and 3D cubic lattice models. However, there are also problems associated with 2D square and 3D cubic lattice models, for instance, the Parity problem and the poor approximation rate. This study focused on the 2D triangular lattice model and proposed an elite-based reproduction strategy (ERS) and genetic algorithm (GA) for protein structure prediction on the 2D triangular lattice in combination with the approach described by Unger and Moult. Our results showed that ERS-UGA can significantly improve UGA and find conformations of lower free energy compared with previous studies.

1 Introduction

A great challenge in computational molecular biology is predicate a protein structure based on its primary amino acid sequence [1]. Since the HP lattice model developed by Lau and Dill [2] and used by others to derive similar approximation and heuristic search algorithms, a variety of lattice models have been proven useful to explore the relationship between the primary amino acid sequence and its native folding structure, particularly in protein folding problems (PFP) and protein structure prediction (PSP). The main purpose of the HP lattice model is to understand the physicochemical principle of protein folding during the modelling process of searching for the lowest free-energy conformation of a protein.

There are two types of commonly used lattice models, namely, 2D square and triangular lattice models and 3D cubic and face-centered-cubic (FCC) lattice models. In 1996, Decatur and Batzoglou [3] pointed out a 'parity' problem in the 2D square and 3D cubic lattice models. Only if two residues are at an even distance from one another can a topological contact be generated. One protein (PDB ID: ICNL) is used in this study as an example. This protein has a short amino acid sequence: GCCSDPRCAWRC. It is known that a topological contact can be generated between 2nd and 12th residues and between 3rd and 8th residues. As the distance between the 3rd and the 8th residues is even (which is 4), it can be shown in the 2D square or 3D cubic lattice models (illustrated by the dashed line in Figure 1). However, the odd distance between 2nd and 12th residues (which is 9 in this case), cannot be shown in the 2D square or 3D cubic lattice models. Figure 1 Illustration of Parity Problem Two residues must have an even topological distance to have a topological contact.

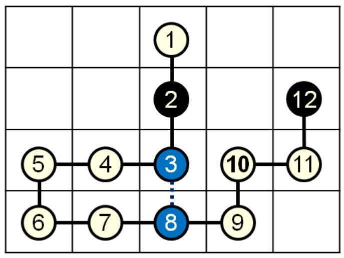

Figure 1. **Parity problem: Two residues must have an even topological distance to have a topological contact.**

In the past, some researchers have compared the approximation ratios of various lattice models as shown in Table I. In the 2D lattice model, a relative higher approximation ratio can be achieved by 2D triangular, while 3D FCC can yield the best approximation ratio in 3D lattice model. However, due to the many benchmarks associated with the square lattice model, the large amount of data accumulated over the years and the availability of comparison with different strategies and modeling methods, many researchers have favored and focused research on the 2D square and 3D cubic lattice models. As a result, little has been done in the direction of the 2D triangular and 3D FCC lattice models.

TABLE I. VARIOUS LATTICE MODEL APPROXIMATION RATIOS

Lattice Type	Approx. ratio	Literature
2D Square	1/4 (25%)	Hart and Istrail (1995) [4] Mauri et al. (1999) [5]
	1/3 (33%)	Newman (2002) [6]
2D Triangular	16/30 (53%)	Decatur and Batzoglou (1996) [3]
	6/11(55%)	Agarwala et al. (1997) [7]
3D cubic	3/8 (38%)	Hart and Istrail (1995) [4]
3D Triangular	16/30 (53%)	Decatur and Batzoglou (1996) [3]
	3/5 (60%)	Agarwala et al. (1997) [7]
3D FCC	31/36 (86%)	Hart and Istrail (1997) [8]
2D hexagonal	1/6 (17%)	Jiang and Zhu (2005) [9]

In the present study with a focus on the 2D triangular lattice models, an elite-based reproduction strategy (ERS) genetic algorithm (GA) was proposed for the protein structure prediction on the 2D triangular lattice. In comparison with previous research by others, our data show that a lower free-energy confirmation can be obtained in the modeling process without the occurrence of Parity problem. Furthermore, the predication approximation to the native physical protein structure (with short sequences) is significantly higher as illustrated in Figure 2.

Figure 2 a) the protein structure established experimentally b) predication of the known protein structure as in a) using 2D triangular lattice. Apparently, the Parity problem is solved between the 2nd and 12th residues as a topological contact can also be generated as between the 3rd and 8th residues.

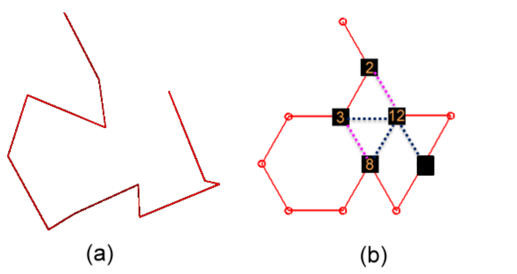

Figure 2. **a) the protein structure established experimentally b) predication of the known protein structure as in a) using 2D triangular lattice.**

The remainder of this paper is structured as follows: Section 2 gives the preliminaries and the formal definition of the protein structure prediction problem in the HP 2D triangular lattice model. Section 3 describes the methodology used in the study. A comparison of the results is presented and discussed in section 4 followed by the conclusion in section 5.

2 Preliminaries

Proteins play fundamental and crucial roles in nearly all biological processes, such as, enzymatic catalysis, signaling transduction, embryonic development, DNA and RNA synthesis and others. It has been a long-standing goal of molecular biology to predict the tertiary structure of a protein from its primary amino acid sequence [10,11].

This paper emphasizes research on ***ab initio*** modeling. The 2D HP triangular lattice model discussed here is one such simplified method and is thought to be the best two-dimensional model in protein structure prediction at present.

A. HP lattice model

The HP lattice model [2] is the most frequently used simplified model, which is based on the observation that the hydrophobic interaction between the amino acid residues is the driving force for the protein folding and for the development of native state in proteins [12]. In this model, each amino acid is classified based on its hydrophobic characteristics as an H (hydrophobic or non-polar) or a P (hydrophilic or polar). The HP lattice model allows HP protein sequences to be configured as self-avoiding walks (SAW) on the lattice path favoring an energy free state due to HH interaction. The energy of a given conformation is defined as the number of topological neighboring (TN) contacts between those Hs, which are not adjacent in the sequence. Figure 3 shows an example for the 2D triangular lattice model.

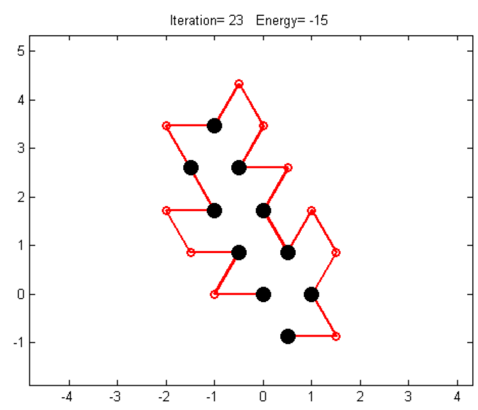

Figure 3. **An optimal conformation for the sequence "(HP)2PH(HP)2(PH)2H P(PH)2" in a 2D triangular lattice model.** The black filled dots denote the hydrophobic amino acid and the white open squares denote the hydrophilic amino acids. The H-H contacts (free energy) in the conformation,

which are assigned the energy value of -1. The free energy is defined as a minimum value; the maximum number of H-H contact is given in the case of two-dimensional models. Figure 2 illustrates a protein structure with 15 H-H contacts (energy= -15).

B. Calculating the free energy

The free energy for the protein can be calculated by using the following formulae [13]:

$$\epsilon_{ij} = \begin{cases} -1.0 & \text{the pair of H and H residues} \\ 0.0 & \text{others} \end{cases} \quad (1)$$

$$E = \sum_{i,j} \Delta r_{ij} \epsilon_{ij}, \quad (2)$$

where the parameter

$$\Delta r_{ij} = \begin{cases} 1 & S_i \text{ and } S_j \text{ are adjacent but not connected amino acids} \\ 0 & \text{others} \end{cases} \quad (3)$$

Hence, the problem of the optimization of protein folding into the calculation of the minimal free energy of the protein folding conformation can be transformed. As a result, the following problem can be formally defined: given an HP sequence $s = s_1, s_2, \ldots, s_n$, find an energy-minimizing conformation of s; that is: find $c^* \in C(s)$ such that $E(c^*) = \min\{E(c) \mid c \in C\}$, where $C(s)$ is the set of all valid conformations for s [14].

C. Triangular lattice model

A significant drawback of the cubic lattice [3] is that if two residues are at any even distance from one another in the primary sequence then they cannot be in topological contact with one another when the protein is embedded in this lattice. For example, on the square lattice, two amino acids in contact in any folding must be an odd distance away in the protein sequence [3]. Joel et al. [15] introduced a 2D triangular lattice model. Consequently, the 2D triangular lattice can be described by a diagram as in Figure 4.

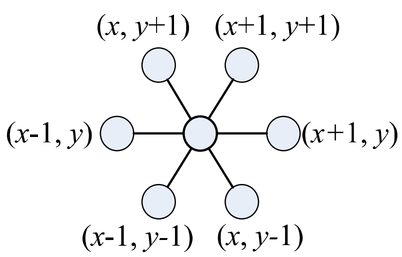

Figure 4. **Neighbors of vertex (x, y)**

In the two-dimensional triangular lattice, each lattice point has six neighbours. Since each residue has two covalent neighbours except the first and the last residues, a residue at a lattice point may be in topological contact with at most four other residues. Thus, each residue may be involved in at most 4 H-H contacts [15].

The unit vectors shown in Figure 3 are logically defined. Real units require normalization by $\sqrt{2}$ and are (1,0),(-1,0), $(-1/2, \frac{\sqrt{3}}{2})$,$(1/2, -\frac{\sqrt{3}}{2})$,$(1/2, \frac{\sqrt{3}}{2})$,$(-1/2, -\frac{\sqrt{3}}{2})$.

After the unit vectors are obtained in the triangular lattice, it is much easier to model protein conformation on a two-dimensional triangular lattice and not exhibit the parity problem [3]. However, the lattice model of protein conformation as a self-avoiding walk is NP-complete [16]. As a result, there are usually numerous likely conformations even for a protein with a short amino acid sequence. To solve this problem, many heuristic search algorithms [17-24] have been developed for a variety of lattice models. The algorithm developed from the present study is discussed in the next section.

3 Methods

The Genetic Algorithm (GA) is a method of optimizing random searching by multiple pathways. In order to replace fixed rules with random calculation in search for proper conformation, GA is chosen as the solution in the PSP problem in our study. Overall, GA gives very robust performance due to its inherent advantages; however, its efficiency in solving the PSP still needs further improvements. There are two key operations in the GA : crossover and mutation. Crossover is capable of linking two different chromosomes very efficiently to form a new conformation, while mutation will not be restricted by local optimum. The limits of HP lattice models, on the other hand, can generate many invalid conformations after operating crossover and mutation, which results in the reduction of the predication efficiency.

A revolving crossover and mutation operator was proposed by Unger and Moult [17] to solve the problem of invalid conformations and significant enhancement has been achieved. Hoque et al.[18] and Böckenhauer et al. [19] are among the few researchers who use 2D triangular lattice model to predicate protein structure. The Hybrid Genetic Algorithm (HGA) as described by Hoque et al. [18] is an approach to optimize the formation of hydrophobic core of a protein, which gives better results than SGA. Böckenhauer et al. [19] described a local search approach that also shows satisfactory protein structure prediction. Böckenhauer et al. [19] used the library of Backofen and Will [25] and extended this library by implementing the 2D triangular lattice and the pull move set for triangular lattice models, which enabled them to obtain (similar) satisfactory results.

In the present study, the approach developed by Unger and Moult [17] was applied on the triangular lattice model in combination with an elite-based reproduction strategy (ERS) [23]. By comparing the results, it is hoped that reasonable improvements can be achieved by the proposed method. Figure 5 shows the flowchart of the proposed ERS-UGA.

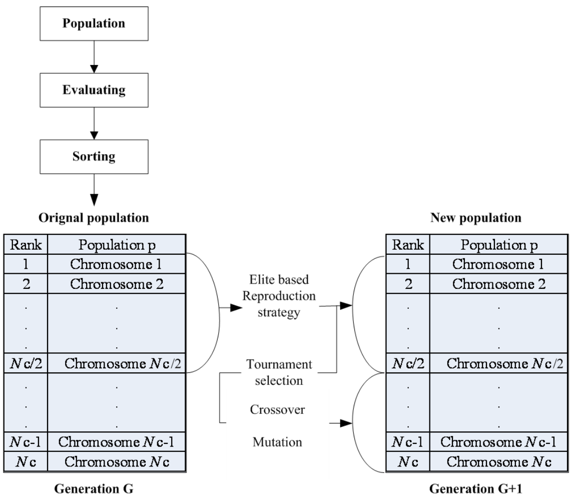

Figure 5. **Flowchart of the proposed ERS.**

A. Initialization step

If the input amino acid sequence is of length n, then each individual in the population is a string of length n - 1 over the symbols = { *L*; *R*; *LU*; *LD* ; *RU*; *RD* }, and that denotes a valid conformation in the 2D triangular lattice [17,20]. The symbols *L*; *R*; *LU*; *LD*; *RU*; and *RD* are used to denote the fold directions: *L* is for left, *R* is for right, *LU* is for left-up, *LD* is for left-down, *RU* is for right-up and *RD* is right-down in the encoding scheme, respectively. An initial population is generated randomly and initializes an n - 1 dimensional space within a fixed range. The population size is set at 100 (sequence 7 is an exception, which has the population (size) set at 500). However, the initialization population size is set as double.

- Evaluating**.** The evaluating step evaluates each chromosome in a population. The goal of the ERS-GA method is to minimize the fitness value. The lower a fitness value, the better the fitness. The fitness function is used by equation (1-3).

- Sorting**.** After the evaluating step, we sort the chromosomes in the population. After the whole population is sorted, the chromosomes are sorted in each group in the top half of the population. This sorting can help to perform the reproduction step because we can keep the best chromosome in each group. After sorting the chromosomes in the population, the algorithm goes to the next step.

B. Elite-Based Reproduction Strategy (ERS)

Reproduction is a process in which individual strings are copied according to their fitness value. In this study, we use the tournament selection method for this reproduction process. In particular, a height elite-based strategy (elite-based rate 50%) is used and the best performing individuals in the top half of the population [23] advance to the next generation. The other half is generated to perform crossover and mutation operations on individuals in the top half of the parent generation.

C. Crossover step

The operation occurred for a selected pair with a crossover rate that was set to 0.8 in this study. The first step is to select the individuals from the population for the crossover. Tournament selection is used to select the top-half of the best performing individuals [23]. The individuals were crossed and separated using the revolving crossover as described by Unger and Moult [17].

D. Mutation step

When the mutation points are selected, the mutation rate is 0.4. Mutation operation used in this study was a revolving mutation proposed by Unger and Moult [17].

E. Termination condition

The algorithm was run for a maximum of 300 iterations. The best member of the population was then returned.

All the parameter settings are listed in Table II.

TABLE II. **THE PARAMETERS SETTING**

Operations/Parameters	Setting
Representation	1-6 integer representative of { L; R; LU; LD ; RU; RD }
Initialization	Random
Population size	100 (sequence 7 is an exception, which has the population (size) set at 500)
Crossover	1-point crossover [17]
Crossover rate	0.8

Number of offspring	Is equal to the size of population size
Mutation	{ L; R; LU; LD ; RU; RD }
Mutation rate	0.4[18]
Parents selection	Tournament selection
Survival selection	$\mu + \lambda$
Termination	300 generations

4 Experiment Resules

Sequences 1 through 8 used in this study were described in Jiang et al. [24]. These sequences have been used as the benchmark for the 2D square HP model as shown in Table III. Sequence 9 in Table III was described in Böckenhauer et al. [19]

TABLE III. THE BENCHMARKS FOR 2D TRIANGULAR LATTICE HP MODEL

Seq.	Len.	Protein Sequence
1	20	$(HP)^2PH(HP)^2(PH)^2HP(PH)^2$
2	24	$H^2P^2(HP^2)^6H^2$
3	25	$P^2HP^2(H^2P^4)^3H^2$
4	36	$P(P^2H^2)^2P^5H^5(H^2P^2)^2P^2H(HP^2)^2$
5	48	$P^2H(P^2H^2)^2P^5H^{10}P^6(H^2P^2)^2HP^2H^5$
6	50	$H^2(PH)^3PH^4PH(P^3H)^2P^4(HP^3)^2HPH^4(PH)^3PH^2$
7	60	$P(PH^3)^2H^5P^3H^{10}PHP^3H^{12}P^4H^6PH^2PHP$
8	64	$H^{12}(PH)^2((P^2H^2)^2P^2H)^3(PH)^2H^{11}$
9	54	$H^2(PH)^4\ H^3P(HP^3)^4P(HP^3)^2HP(H)^4(PH)^4H$

The approach used by Unger and Moult [17] was performed in this study with or without combination with an elite-based reproduction strategy (ERS) [23]. The comparison after operating for more than 10 times is shown in Table III. The format of column entries is "Average/Minimum". From Table IV, it is seen that the combination of ERS with UGA gave better results in both average free energy and lowest free energy than using UGA alone exception in sequence 2. However, the optimal energy of these benchmark sequences in the 2D triangular HP model was unknown; nevertheless, when comparing the previous results from others with our own, the ERS-UGA proposed in our study could obtain the same free energy for sequences 1-4. For sequences 5-9, conformations of much lower free energy could be predicated for sequences 5-9 in addition to obtaining the same quantity of free energy as shown in Figure 6.

TABLE IV. COMPARISON OF ERS-UGA APPROACH WITH UGA, SGA, HYBRID GA (HGA), AND THE TABU SEARCH (TS). THE FORMAT OF COLUMN ENTRIES IS 'AVERAGE / MINIMUM'. FIGURES IN BOLD INDICATE THE LOWEST ENERGY.

Seq.	Len.	ERS-UGA	UGA	SGA [18]	HGA [18]	TS [19]	**Conformation**
1	20	-14.7/**-15**	-14.9/**-15**	-/-11	**-/-15**	**-/-15**	Fig.6 (a)
2	24	-14.7/**-17**	-15.2/-16	-/-10	-/-13	**-/-17**	Fig. 6 (b)
3	25	-11.6/**-12**	-11.4/**-12**	-/-10	-/-10	**-/-12**	Fig. 6 (c)
4	36	-21.5/**-24**	-20.6/-22	-/-16	-/-19	**-/-24**	Fig. 6 (d)
5	48	-38.1/**-42**	-36.6/-39	-/-26	-/-32	-/-40	Fig. 6 (e)
6	50	**-34.9/-37**	-34.2/**-37**	-/-21	-/-23	-/-	Fig. 6 (f)
7	60	-66.5/**-70**	-61.5/-62	-/-40	-/-46	**-/-70**	Fig. 6 (g)
8	64	**-56/-60**	-54.5/-59	-/-33	-/-46	-/-50	Fig. 6 (h)
9	54	**-35.3/-39**	-34.8/-37	-/-	-/-	-/-31	Fig. 6 (i)

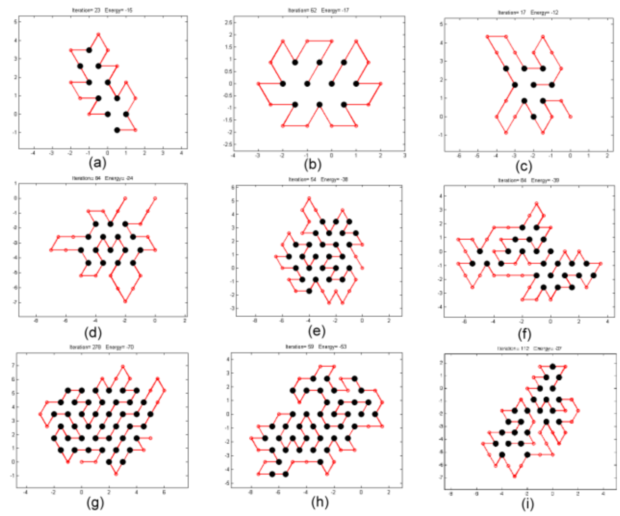

Figure 6. **(a) to (i) Results of the structure of 9 protein sequences**

5 Conclusion

In the ***ab initio*** technique, the lattice model is one of the most frequently used methods in protein structure prediction. Numerous lattice models have been proposed in the past, with 2D square and 3D cubic lattice models as the most preferred by a wide range of researchers. However, when it is necessary to improve the approximation ratio of the computational predication to native physical structure of a protein, different lattice models such as 2D triangular and 3D FCC lattice models have to be selected.

A very important reason for using these two different models is that they could achieve high approximation ratio of 53% [3] and 86%[8]

without the Parity problem [3]. As a result, an elite-based reproduction strategy (ERS) genetic algorithm (GA) was proposed for protein structure prediction on the 2D triangular lattice in combination with the method by Unger and Moult [17]. From our study, it can be concluded that the ERS-UGA approach is capable of finding conformations of lower free energy in addition to improving UGA itself in comparison to previous studies.

References

[1] P. Bradley, K. M. S. Misura, and D. Baker, "**Prediction for Small Proteins Toward High-Resolution de Novo Structure**", *Science* 2005, 309, pp.1868

[2] K. F. Lau and K. A. Dill, "**lattice statistical mechanics model of the conformation and sequence space of proteins**", *Macromolecules*, 1989, pp. 3986-3997.

[3] S. Decatur and S. Batzoglou, "**Protein folding in the Hydrophobic-Polar model on the 3D triangular lattice**", *In 6th Annual MIT Laboratory for Computer Science Student Workshop on Computing Technologies*, 1996.

[4] W. E. Hart and S. Istrail, "**Fast protein folding in the Hydrophobic-Hydrophilic model within three-eighths of optimal (extended abstract)**", *Proceedings of 27th Annual ACM Symposium on Theory of Computation (STOC95)*, 1995, pp. 157–168.

[5] Mauri, G. Pavesi and A. Piccolboni, "**Approximation algorithms for protein folding prediction**", in Proc. *10th Annual ACM-SIAM Symposium on Discrete Algorithms* (SODA'99), 1999, pp. 945–946.

[6] Newman, "**A new algorithm for protein folding in the HP model**", Proceedings of the *13th ACM-SIAM Symposium on Discrete Algorithms*, SODA, 2002, pp. 876–884.

[7] R. Agarwala, S. Batzogloa, V. Dancik, S. E. Decatur, S. Hannenhalli, M. Farach, S. Muthukrishnan, and S. Skiena, "**Local rules for protein folding on a triangular lattice and generalized hydrophobicity in the hp model**", *Journal of Computational Biology*, 1997, pp. 276–296.

[8] W.E. Hart and S. Istrail, "**Lattice and Off-Lattice Side Chain Models of Protein Folding: Linear Time Structure Prediction Better than 86% of Optimal**", *Journal of Computational Biology*, 1997, pp.241–259.

[9] M. Jiang and B. Zhu, "**Protein folding in the hexagonal lattice in the hp model**", J. of Bioinformatics and Computational Biology, 2005, pp. 19–34.

[10] E. Mirsky, and L. Pauling, "**On the structure of native, denatured and coagulated proteins**", *Proc. Natl. Acad. Sci.* USA, 1936, pp. 439-447.

[11] C.A. Orengo, and A.E. Todd , "**From protein structure to function**", *Curr. Opin. Struct. Biol.*, 1999, pp.374-382.

[12] Y-Z. Guoa, E-M. Fenga, and Y. Wangb, "**Optimal HP configurations of proteins by combining local search with elastic net algorithm**", *Journal of Biochemical and Biophysical Methods*, 2007, pp. 335-340.

[13] C. Huang, X. Yang, and Z. He, "**Protein folding simulations of 2D HP model by the genetic algorithm based on optimal secondary structures**", *Computational Biology and Chemistry*, pp. 137–142, 2010

[14] Shmygelska and H. H. Hoos, "**An ant colony optimisation algorithm for the 2D and 3D hydrophobic polar protein folding problem**", *BMC Bioinformatics*, pp. 30, 2005

[15] Joel, M. Martin, and J. Minghui, "**RNA folding on the 3D triangular lattice**", *BMC Bioinformatics*, 2009, doi:10.1186/1471-2105-10-369.

[16] P. Crescenzi, D. Goldman, C. Papadimitriou, A. Piccolboni, and M. Yannakakis, "**On the complexity of protein folding**", *J. Comp. Biol.*, 1998, pp. 409–422.

[17] R. Unger, and J. Moult, "**Genetic algorithms for protein folding simulations**", *Journal of Molecular Biology*, 1993, pp. 75–81.

[18] M.T. Hoque, M. Chetty, and L.S. Dooley, "**A hybrid genetic algorithm for 2D FCC hydrophobic–hydrophilic lattice model to predict protein folding**", *in: Proceedings of the 19th ACS Australian Joint Conference on Artificial Intelligence*, LNAI, Springer, 2006, pp. 867–876.

[19] H-J. Böckenhauer, A. D. Ullah, L. Kapsokalivas, and K. Steinhöfel, "**A Local Move Set for Protein Folding in Triangular Lattice Models**", *Algorithms in Bioinformatics*, LNCS, 2008, pp. 369-381.

[20] A. Albrechta, A. Skaliotisb, and K. Steinhöfelb, "**Stochastic protein folding simulation in the three-dimensional HP-model**", *Computational Biology and Chemistry*, 2008, pp. 248-255.

[21] Dayem Ullah, L. Kapsokalivas, M. Mann, and K. Steinhöfel, "**Protein Folding Simulation by Two-Stage Optimization**", *In Proc. of ISICA'09*, Wuhan, China,2009, pages 138-145.

[22] X. Zhao, "**Advances on protein folding simulations based on the lattice HP models with natural computing**", *Applied Soft Computing*, 2008, pp. 1029-1040

[23] J. Lin, and Y. C. Hsu, "**Reinforcement hybrid evolutionary learning for recurrent wavelet-based neuro-fuzzy systems**", *IEEE Transactions on Fuzzy Systems*, 2007, pp. 729–745.

[24] T. Jiang, Q. Cui, G. Shi, and S. Ma, "**Protein folding simulations for the hydrophobic-hydrophilic model by combining tabu search with genetic algorithms**", *Journal of Chemical Physics*, 2003, pp.4592-4596.

[25] R. Backofen, and S.Will, "**A constraint-based approach to fast and exact structure prediction in three-dimensional protein models**", Constraints, 2006, pp. 5–30